

Finding Unit Rates

Unit rates from two quantities, ratio tables, comparing rates, and applying a unit rate

Directions:

- Every problem comes with a ratio table. Fill in the “per one” column first — that column *is* the unit rate.
- Write the division you used beside the table, then write the rate with its units on the answer line.
- A unit rate always answers the question “how much for exactly one?”

1. A box of 8 granola bars costs 24 dollars. What is the cost per bar?

Bars	8	1
Cost (dollars)	24	

Answer: _____ dollars per bar

2. A printer produces 90 pages in 6 minutes at a steady speed. How many pages does it print in one minute?

Minutes	6	1
Pages	90	

Answer: _____ pages per minute

3. The ticket table below is missing two entries. Complete the whole table, then state the unit rate.

Tickets	4	8	1
Cost (dollars)	52		

Answer: _____ dollars per ticket

4. A bus covers 156 miles in 3 hours at a steady speed. Complete the table and give the speed as a unit rate.

Hours	3	2	1
Miles	156		

Answer: _____ miles per hour

5. Store A sells cashews at 5 pounds for 20 dollars. Store B sells the same cashews at 8 pounds for 28 dollars. Work out both prices per pound, decide which store is cheaper, and write *that* store's rate on the answer line.

Store	Pounds	Dollars	Dollars per pound
A	5	20	
B	8	28	

Answer: _____ dollars per pound

6. A recipe makes 12 muffins using 3 cups of flour. Find the muffins-per-cup rate first, then use it: how many muffins can be made with 7 cups of flour?

Cups of flour	3	1	7
Muffins	12		

Answer: _____ muffins

7. Maya runs 12 laps in 4 minutes. Dev runs 21 laps in 7 minutes. Find each runner's laps per minute, then subtract Dev's rate from Maya's rate. Write that difference on the answer line and say in one sentence what it tells you about who is faster.

Runner	Laps	Minutes	Laps per minute
Maya	12	4	
Dev	21	7	

Answer: _____

8. A school orders crayons in identical boxes. Six boxes hold 138 crayons altogether. Complete the table and state the unit rate.

Boxes	6	1	4
Crayons	138		

Answer: _____ crayons per box

- 9.** A pump moves 84 gallons of water in 12 minutes at a steady rate. Find the gallons-per-minute rate, then work out how long the pump needs to move 210 gallons.

Minutes	12	1	
Gallons	84		210

Answer: _____ minutes

- 10.** A 3-pack of sports drink costs 18 dollars; a 5-pack costs 25 dollars. Jae says the 3-pack is the better buy “because 18 is less than 25”. Explain in one sentence why that reasoning is not enough, then give the price per bottle of the pack that really is the better buy.

Pack	Bottles	Dollars	Dollars per bottle
Small	3	18	
Large	5	25	

Answer: _____ dollars per bottle